

Model Name	Underlying LM	Vision Encoder	Size	HuggingFace Path
LLaVA-1.5	Vicuna	CLIP (ViT-L/14)	7B	llava-hf/llava-1.5-7b-hf
LLaVA-Next (LLaVA-1.6)	Mistral	CLIP (ViT-L/14)	7B	llava-hf/llava-v1.6-mistral-7b-hf
LLaVA-OneVision	Qwen-2	SigLIP	7B	llava-hf/llava-onevision-qwen2-7b-ov-hf
Qwen2-VL	Qwen2	DFN-ViT w/ RoPE-2D	7B	Qwen/Qwen2-VL-7B-Instruct
InternVL	QLLaMA	InternViT	8B	OpenGVLab/InternVL2-8B
Molmo	OLMo	MetaCLIP	7B	cyan2k/molmo-7B-D-bnb-4bit
Janus Pro	DeepSeek-LLM	SigLIP-Large-Patch16-384	7B	deepseek-ai/Janus-Pro-7B
Math-LLaVA	Vicuna	CLIP (ViT-L/14)	13B	Zhiqiang007/Math-LLaVA
G-LLaVA	Vicuna	CLIP (ViT-L/14)	7B	renjiepi/G-LLaVA-7B
Math-PUMA	Qwen2	DFN-ViT w/ RoPE-2D	7B	Math-PUMA/Math-PUMA_Qwen2VL-7B

Table 8: Details of evaluated models, including underlying language models, vision encoders, sizes, and Hugging Face model paths.

distinct shape color chosen from 20 hues. The polygon vertices are generated with constraints to ensure reasonable side lengths and angles, avoiding excessively sharp or flat corners. The number of sides is correctly recorded in the filenames to ensure metadata accuracy. In total, we generate **200** unique irregular polygons for analysis.

Common Shapes. We consider four shape configurations: plus sign (or a cross), star, arrow, and arrow on plus sign (or a triangle on top of a cross). Each shape undergoes a uniform scaling transformation and is randomly rotated between 0° and 360° . This process results in a dataset of **200** uniquely transformed composite shapes.

C What does the text-decoder know?

Name the $\langle n \rangle$ -sided polygon / Num of sides in $\langle shape \rangle$	
LM Only	Accuracy (%)
LLaVA-1.5	67% / 100%
LLaVA-Next	100% / 100%
LLaVA-OneVision	100% / 100%
Qwen2-VL	100% / 100%
LLaMA-3.2	100% / 100%
InternVL	100% / 100%
Molmo	100% / 100%
Janus-Pro	100% / 100%
Math-LLaVA	100% / 100%
G-LLaVA	50% / 100%
Math-PUMA	100% / 100%
GPT-4o	100% / 100%
GPT-4-Turbo	100% / 100%

Table 9: Accuracy of the LLM-backbone on the polygon naming and side counting tasks: “What is the name of a $\langle n \rangle$ -sided regular polygon? / How many sides does $\langle shape \rangle$ have?” for $n \in \{3, 4, \dots, 8\}$ and shape in $\{\text{triangle, square, } \dots, \text{octagon}\}$.

To evaluate LLMs’ recall of basic geometric properties, we pose two sets of six questions: (1) naming a polygon given its number of sides (3–8)

and (2) identifying the number of sides in a named shape. As shown in Appendix Table 9, models perform nearly perfectly, with all achieving 100% accuracy on the second task and 11 out of 13 models doing so on the first. The exceptions are LLaVA-1.5, which misclassified a heptagon as a hexagon and a square as a tetrahedron, and G-LLaVA, which answered “regular polygon” instead of the specific shape name for several cases.

D What does the vision-encoder see?

Figure 8 presents t-SNE visualizations of vision encoder embeddings across all models. Common shapes, such as triangles and squares, consistently form distinct clusters, whereas less frequent shapes, including hexagons, heptagons, and octagons, exhibit dispersed and overlapping embeddings. This confirms that vision encoders struggle to differentiate between less common shapes, leading to poor shape generalization. Nearest neighbor analysis, in Table 10, further supports this, showing that models reliably group common shapes but misclassify rarer ones, often embedding them into mixed clusters.

E Google N-Grams Analysis of Shape Frequencies

Figure 9 presents an analysis of shape mention frequencies over time using data from Google Books N-Grams. The plot highlights the relative prominence of different shapes and their corresponding numbers in written text. Notably, triangles and squares are mentioned significantly more frequently than other shapes, reflecting a clear bias in natural language usage.

This bias in textual representation may contribute to discrepancies in model performance, as models trained on large language corpora are more likely to encounter these common shapes during training. For instance, “heptagon,” the only shape

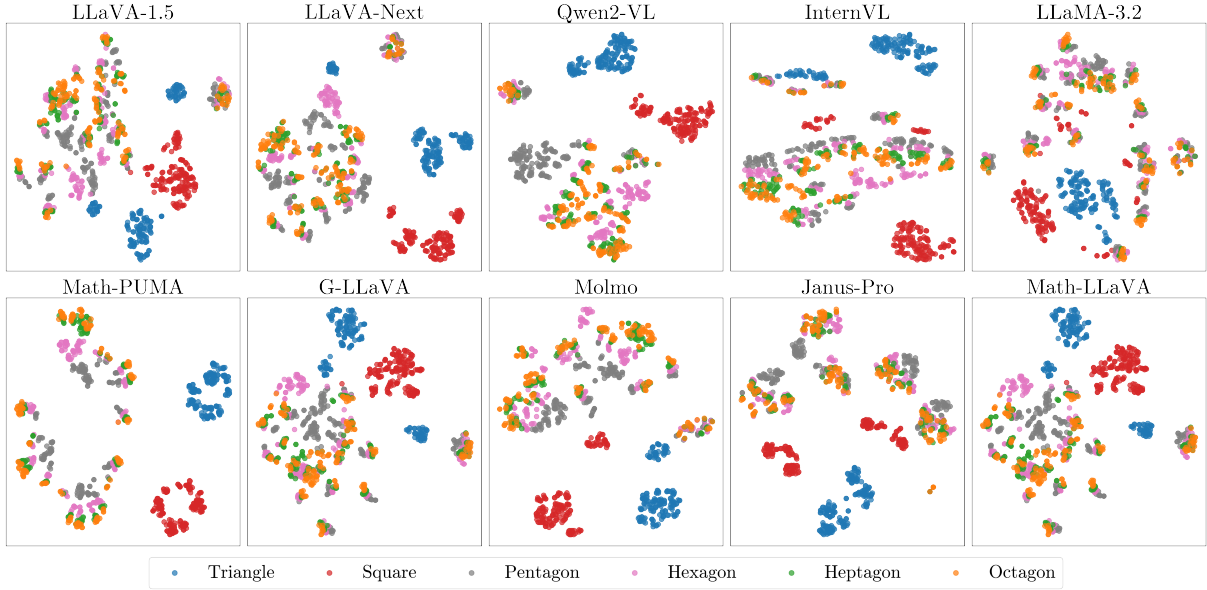


Figure 8: t-SNE plots from the vision encoders of all models considered in this paper. For each image, we take the average over all image patches.

Model	Triangle	Square	Pentagon	Hexagon	Heptagon	Octagon
LLaVA-1.5	95%	95%	65%	50%	44%	48%
LLaVA-Next	99%	97%	66%	58%	50%	46%
Qwen2-VL	100%	98%	84%	67%	51%	50%
InternVL	97%	90%	58%	50%	47%	47%
LLaVA-OneVision	100%	100%	82%	58%	52%	52%
LLaMA-3.2	87%	79%	40%	37%	36%	34%
Math-PUMA	100%	100%	76%	57%	51%	51%
G-LLaVA	96%	97%	65%	48%	44%	47%
Molmo	99%	97%	76%	54%	51%	46%
Janus-Pro	98%	95%	74%	47%	46%	43%
Math-LLaVA	96%	97%	65%	48%	44%	48%

Table 10: Average percentage of nearest neighbors of vision encoder embeddings that share the same shape label.

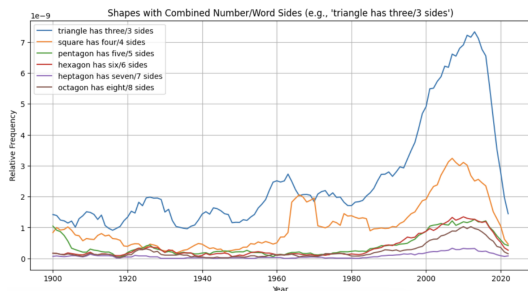


Figure 9: Google Books N-Grams showing frequency of different shape mentions with their corresponding numbers over the years. There is a clear bias towards triangles and squares compared to the rest. ‘Heptagon’ is the only shape GPT-4-Turbo misclassified and it is the least frequent in this plot.

misclassified by GPT-4-Turbo in our experiments, is also the least frequent shape in this plot. This suggests a possible correlation between linguistic frequency and model performance on geometric reasoning tasks.

F Molmo Demo Examples

To evaluate the pointing ability of Molmo, we experiment using the HuggingFace-hosted model as well as its interactive demo website. Specifically, we prompt the model three different times with variations of prompts closely matching examples in the Molmo paper. The task involves pointing to and identifying the sides of a simple black heptagon.

In all prompt variations, Molmo consistently points to the heptagon but incorrectly assigns the points, resulting in 10 sides instead of the correct 7. However, when the pointing task is omitted, and the prompt is modified to include the instruction “An-

	Direct	MathVerse CoT	VC-CoT
LLaVA-1.5	29.39%	29.97%	28.34%
LLaVA-Next	28.48%	23.46%	30.95%
LLaVA-OneVision	47.88%	48.62%	50.16%
Qwen2-VL	35.76%	41.11%	41.95%
InternVL	34.85%	47.09%	44.86%
LLaMA-3.2	22.73%	32.72%	35.20%
Math-LLaVA	36.06%	34.86%	36.14%
G-LLaVA	35.76%	36.70%	35.83%
Math-PUMA	39.70%	36.09%	37.35%
Average	34.51%	36.73%	37.86%

Table 11: Impact of Visually-Cued CoT (VC-CoT) on MathVerse across different models. Compared to standard CoT prompting, explicitly guiding models to extract and process visual information leads to consistent performance improvements. This table includes all models not in Table 7.

swer with the number”, Molmo correctly identifies the number of sides as 7 (which correspond to the results in Table 1). This suggests that while Molmo performs reasonably well in direct counting tasks, its pointing mechanism struggles to accurately map geometric features in images.

Figure 10 shows real screenshots taken from the Molmo demo website, illustrating its outputs for the shape-recognition and side-counting task.

G Visually-Cued Chain-of-Thought Shape Results

Tables 12 and 13 present results for the remaining models not included in Table 6.

Prior work has shown that standard Chain-of-Thought (CoT) prompting often fails to improve, and in some cases, even decreases mathematical reasoning performance in MLLMs (Wang et al., 2024a). This has been attributed to the lack of fine-grained visual information captured by most vision encoders, which struggle to integrate step-by-step reasoning with visual context (Zhang et al., 2024c,b). Specifically, models trained on vision-language instruction datasets tend to prioritize short-answer responses, limiting their ability to perform multi-step reasoning (Zhang et al., 2024c).

Our findings reinforce some of these observations. As seen in Table 12, most models struggle with side-counting on a heptagon, even with explicit ordered labels (1-7). VC-CoT, helps promote some correct predictions, but not in all models. We hypothesize that their rigid clustering patterns (Figure 2) prevent them from adapting their reasoning when prompted. For example, LLaVA-OneVision misclassified a heptagon as a hexagon,

stating: “The shape is a hexagon. The letters are A, B, C, D, E, F, and G. There are 7 letters in total... A hexagon has 6 sides.” These models fail to adjust their predictions even when provided with explicit step-by-step reasoning.

As shown in Table 13, some models exhibit different behaviors when reasoning about more complex composite shapes, such as the “arrow on plus sign” configuration. Notably, Math-PUMA’s strong visual priors make VC-CoT ineffective for counting heptagon sides, where deeply ingrained biases interfere with reasoning. However, on far more complex shapes, like a 15-sided “arrow on plus sign,” VC-CoT significantly improves performance. This is likely because the lack of a strong prior allows Math-PUMA to engage in step-by-step reasoning rather than relying on memorized heuristics.

H VC-CoT Results on MathVerse

Table 11 presents the impact of Visually-Cued Chain-of-Thought (VC-CoT) prompting on MathVerse accuracy across models not included in Table 7. Prior work has demonstrated that standard Chain-of-Thought (CoT) prompting can be ineffective, and in some cases, even detrimental to mathematical reasoning performance in multimodal large language models (MLLMs) (Wang et al., 2024a). This is largely due to the training biases of vision-language models, which tend to favor short-answer responses over structured, multi-step reasoning (Zhang et al., 2024c).

Despite these challenges, our results show that VC-CoT improves overall performance across these models, with an average accuracy increase from 34.51% (Direct) to 37.86% (VC-CoT). Notably, 9 out of 13 models (including ones in Table 7) benefit from visually guided reasoning, suggesting that explicit extraction and processing of visual information can help compensate for the weaknesses of generic CoT prompting.

These findings suggest that while VC-CoT is not a universal solution, it provides a tangible benefit even for models not explicitly designed for reasoning through CoT. Future work may explore fine-tuning strategies that further leverage visual embeddings to enhance mathematical problem-solving capabilities in MLLMs.